

VOLUMETRIC OPTICAL PROCESSING UNIT (VOPU)

Executive Briefing: Physics-First Analysis of Monolithic 3D Photonic AI Hardware

Date: September 2026 | Baseline Status: Baseline Simulation Report (Revised) | Evidence Level: S / D Grounded

EXECUTIVE SUMMARY & STRATEGIC PREMISE

Key Takeaway: VOPU shifts optical AI computing from planar integrated circuits to monolithic 3D solid substrates. Baseline physics simulations confirm that a **5 to 10-node linear demonstrator is feasible**. Waveguide propagation loss is negligible (~ 0.011 dB/cm, behaving as single-mode optical fiber). The primary binding constraint on network depth is **phase error accumulation (± 12 nm fab tolerance $\rightarrow 16.4^\circ/\text{node } \sigma$)**, not optical power loss. Terminal-node directed scattering acts as active weight multiplication, while path integral summation reconstructs the inferred output.

1. Architectural Paradigm & Core Physics

Standard photonic integrated circuits (PICs) rely on 2D planar silicon or Thin-Film Lithium Niobate (TFLN) waveguides, where 2D waveguide crossings, insertion losses, and spatial routing congestion create severe scaling bottlenecks. The **Volumetric Optical Processing Unit (VOPU)** resolves these topological limits by compiling neural network state dictionaries directly into a monolithic 3D optical crystal written via sub-diffraction **Implosion Carving (ImpCarv)**.

Architectural Component	Physical Mechanism	System Function & Provenance
Directed Scattering	Terminal-node geometry & index contrast	Terminal nodes scatter light in programmed patterns implementing complex weights. Directed scattering IS weight multiplication.
Fiber-Optic Propagation	3D single-mode hydrogel channels	Waveguide channels act as optical fibers. Propagation loss is ~ 0.011 dB/cm (negligible over cm scales).
Path Integral Summation	Coherent field optical interference	Scattered fields from all nodes interfere at output facets. Interference patterns constitute the inferred output.
Epsilon-Thresholding (ϵ)	Reverse-Saturable Absorbers (PbPc)	Saturable material zones act as intensity activation gates (transparent below threshold, absorbing above).
Lambda-Decay (λ)	Chromophore excited-state lifetime ($\tau \approx 1.2$ ns)	Transient refractive-index relaxation provides short-term temporal memory across sequential optical pulses.
Cross-Phase Modulation	Nonlinear Kerr interaction (XPM)	Overlapping Query (Q) and Key (K) fields modulate local refractive paths for dynamic operand interaction.

Enabling Fabrication Premise: Peer-reviewed literature (Yang et al., *Nature Photonics*, 2026) demonstrates that ImpCarv photopatterns vacancies in swollen polyacrylate hydrogels down to 67 ± 12 nm lateral width and 22 ± 2 nm axial steps. Isotropic linear shrinkage factors of **5.03 $\times$ (MSF formulation)** to **13.18 $\times$ (HSF formulation)** enable sub-wavelength refractive-index structuring ($\Delta n \approx 0.5$) at visible wavelengths (532 nm).

Earlier architectural specifications incorporated a conservative placeholder attenuation of 0.25 dB/cm. Revised baseline physics simulations establish the **Fiber-Optic Principle**: straight waveguide channels in transparent hydrogel behave with fiber-like efficiency. Commercial PMMA plastic optical fiber achieves 0.001 dB/cm at 520 nm; adding ImpCarv surface roughness scattering ($\sigma = 12$ nm) yields a realistic total attenuation of **0.011 dB/cm** (a 23× reduction in propagation loss).

VOPU Baseline Signal Loss Breakdown at 10-Node Depth (Total: 1.4 dB)

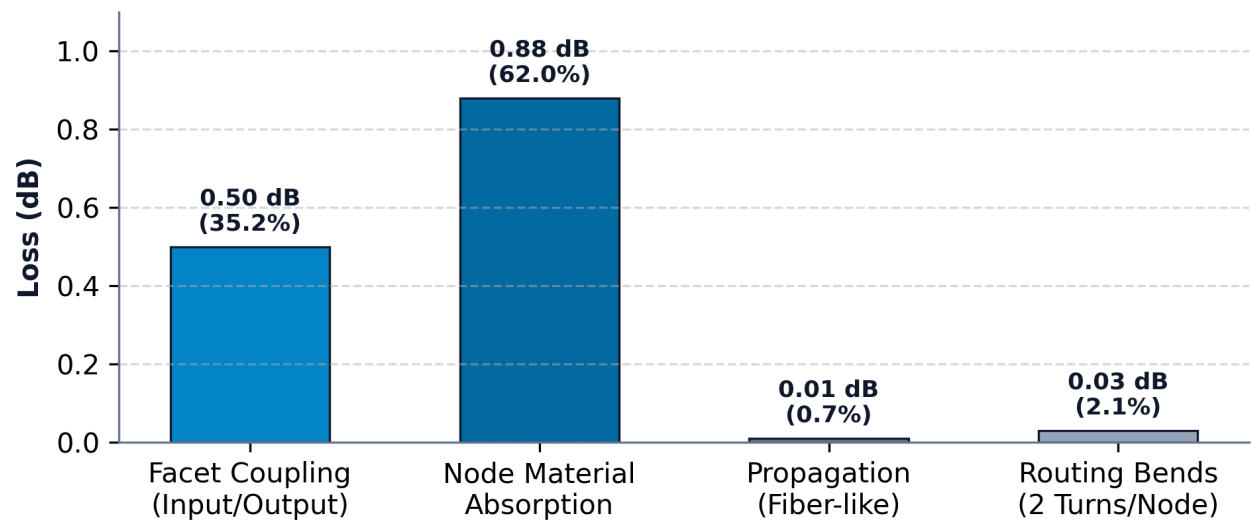


Figure 1: Quantitative signal loss budget breakdown for a 10-node VOPU cascade (Total Loss = 1.4 dB, SNR = 78.6 dB).

Loss Mechanism	Loss Value (10 Nodes)	Share (%)	Physical Origin & Engineering Reality
Facet Coupling	0.50 dB	35.2%	Input/output interface reflections (largest single operational loss term).
Node Material Absorption	0.88 dB	62.0%	Unavoidable absorption during terminal scattering (computation, not waste).
Propagation Attenuation	0.01 dB	0.7%	Fiber-like propagation through hydrogel (negligible engineering concern).
Routing Bends	0.03 dB	2.1%	Low radiation loss across Manhattan-style 90° routing turns.

POWER LOSS IS NON-LIMITING

Cascability Implication: Purely from an optical power perspective, a VOPU volume can support cascade depths of **700 to 1,900+ nodes** before hitting signal-to-noise detection limits. Power loss is *not* the binding constraint on system depth.

3. Primary Engineering Bottlenecks & Risk Register

Because VOPU relies on coherent path integral summation, **phase precision—not optical power—is the binding constraint**. Microscopic fabrication tolerances accumulate phase variance across cascading layers, destroying wave interference fidelity.

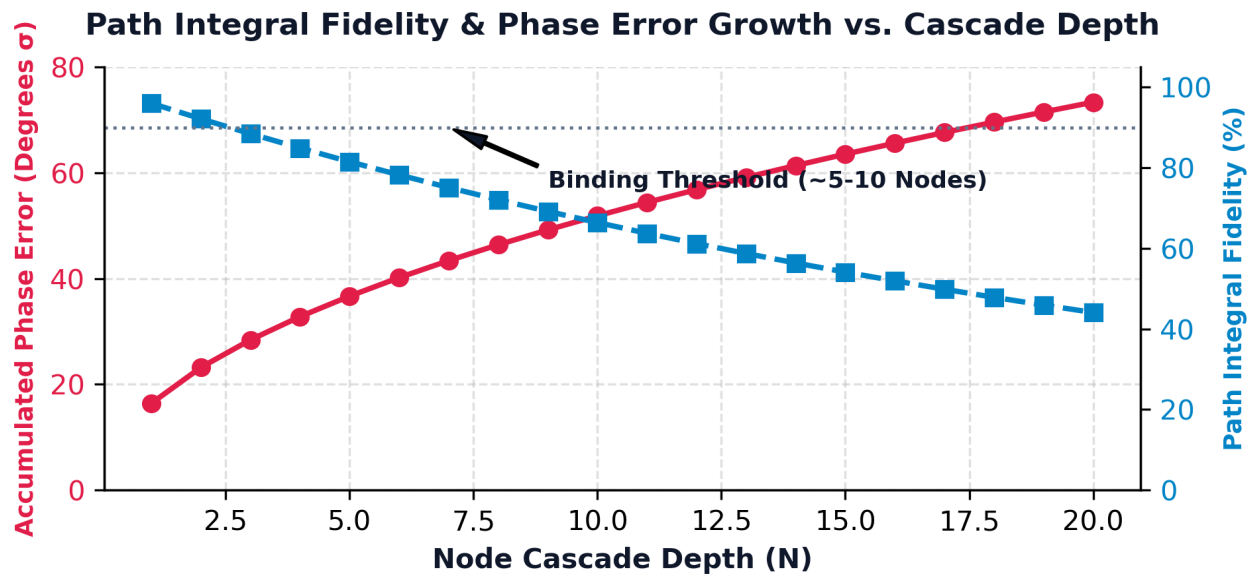


Figure 2: Phase error growth ($\sigma \propto \sqrt{N}$) and path integral fidelity decay versus cascade depth.

Risk / Bottleneck	Severity & Value	Impact on System	Mitigation Strategy
Fabrication Phase Error	13.7° / node (±12 nm fab)	Accumulates as $\sqrt{N}$. Total $\sigma = 16.4^\circ/\text{node}$. Limits depth to 5–10 nodes for 90% fidelity.	Inverse-phase pre-compensation in compiler; active per-node thermal phase calibration.
XPM Kerr Power Gap	13 W (un-doped PMMA)	Pure PMMA Kerr shift requires excessive power for 45° Q/K interaction.	Dope hydrogel with organic chromophores or nanoparticles ($n_2 \sim 10^{-11} \text{ m}^2/\text{W}$) to drop power to ~130 mW.
RSA Activation Drift	17 mW (PbPc @ 532 nm)	Saturable absorber degradation or thermal shift alters activation thresholds.	Characterize Lead Phthalocyanine (PbPc) in solid hydrogel host under E05 protocol.
Compiler-Fab Mismatch	5.03× Shrinkage MSF	Anisotropic shrinkage distorts written scattering geometry and phases.	Preserve inverse deformation tensor map in weight_compiler.py; closed-loop metrology.

4. Competitive Matrix & Strategic Positioning

VOPU occupies a unique position in the optical computing landscape. While commercial competitors adapt 2D manufacturing or free-space diffractive optics, VOPU embeds the full tensor Hamiltonian into a monolithic 3D solid.

Company / System	Architecture Type	Core Tech / Substrate	Strategic Limitations vs. VOPU
Q.ANT (Germany)	Native Photonic NPU	Thin-Film Lithium Niobate (TFLN)	Planar 2D limits; restricted by waveguide crossing congestion and 2D scaling.
Lumai (UK)	3D Optical Matrix Multiplier	Free-space spatial light modulators	Requires discrete optical components and precise free-space alignment.
Peking Univ. (China)	Analog Matrix Chip	Resistive memory / analog circuits	Subject to electronic resistance scaling and thermal dissipation limits.
VOPU (CaelumSculptoris)	Monolithic 3D Photonic Volume	Implosion Carved 3D Hydrogel	Bound 3D volume eliminates crossings; constrained strictly by 3D phase precision.

PROVENANCE & TIMELINE PRECEDENCE

Historical Provenance: Repository records show that the core VOPU concept—mapping LLM weights to 3D waveguide lattices, refractive indices, and node incidence angles—was formulated in late 2024 and December 2025. The subsequent publication of the MIT ImpCarv paper in May 2026 provided the independent, peer-reviewed fabrication proof of concept.

5. Actionable Roadmap & Gated Milestones

Milestone	Work Package	Objective & Experimental Protocol	Go/No-Go Exit Criterion
M1: Linear Path	WP2 — Guide Modeling	Measure straight 3D guide transparency and mode profile under Protocol E01.	Propagation loss < 0.05 dB/cm; single-mode profile confirmed.

Milestone	Work Package	Objective & Experimental Protocol	Go/No-Go Exit Criterion
M2: Node Scattering	WP2 — Scattering Primitive	Calibrate single terminal-node directed scattering pattern under Protocol E03.	Repeatable complex transfer function; scattering efficiency characterized.
M3: Path Integral	WP3 — Path Summation	Test 5-10 node multi-path interference under Protocol E04.	Summed output matches numerical reference within predefined phase budget.
M4: Nonlinearity	WP4 — Material Characterization	Characterize PbPc RSA thresholding (E05) and Kerr XPM shift (E07) in hydrogel.	Repeatable non-damaging activation threshold with quantified power window.
M5: Primitive Demo	WP6 — Inference Primitive	Execute compiled matrix-vector product on physical 3D core.	Optical inference output meets numerical reference accuracy target.

Summary Recommendation: Proceed directly to Milestone M1 (Protocol E01) and M2 (Protocol E03) linear routing validation. Development of deeper networks should be gated on achieving per-node phase error calibration below 10°.